Mathematics - Undergraduate Level Assessment 20 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. Statement 1 | For any two groups G and G', there exists a homomorphism of G into G'. Statement 2 | Every homomorphism is a one-to-one map.
   (a) True, True
   (b) False, False
   (c) True, False
   (d) False, True

2. If A is a subset of the real line R and A contains each rational number, which of the following must be true?
   (a) If A is open, then A = R.
   (b) If A is closed, then A = R.
   (c) If A is uncountable, then A = R.
   (d) If A is uncountable, then A is open.

3. The element (4, 2) of Z__12 x Z__8 has order
   (a) 4
   (b) 8
   (c) 12
   (d) 6

4. Suppose that f(1 + x) = f(x) for all real x. If f is a polynomial and f(5) = 11, then f(15/2)
   (a) -11
   (b) 0
   (c) 11
   (d) 33/2

5. For which of the following rings is it possible for the product of two nonzero elements to be zero?
   (a) The ring of complex numbers
   (b) The ring of integers modulo 11
   (c) The ring of continuous real-valued functions on [0, 1]
   (d) The ring {a + b * sqrt(2) : a and b are rational numbers}

True / False (10 marks)

Answer True or False

6. True or False: The order of the factor group (Z_11 x Z_15)/(<1, 1>) is 1.
7. True or False: The element 2 is a generator for the finite field Z_11, meaning it can produce all non-zero elements of the field through exponentiation.
8. True or False: The characteristic of the ring Z_3 x Z_3 is 3, meaning that adding any element to itself three times results in the zero element.
9. True or False: The dimension of the intersection of two 2-dimensional subspaces of R^4 can be 0, 1, or 2.
10. True or False: The minimum value of the expression x + 4 z, subject to the constraint x^2 + y^2 + z^2 <= 2, is -$\surd$ 34.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Divide the product of the first five positive composite integers by the product of the next five composite integers. Express your answer as a common fraction.
12. Consider sets S, T, and U with functions f: S -> T and g: T -> U such that g f: S -> U is injective. Discuss the implications of this condition on the function f and justify your reasoning.

End of Paper